JUNIK BAE

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EDUCATION

Stanford University Incoming, Sep 2026
Master of Science in Computer Science *Stanford, CA, USA*

Seoul National University March 2019 - Present
Bachelor of Computer Science and Engineering *Seoul, South Korea*
GPA: 4.08 / 4.30 (Cumulative), 4.14 / 4.30 (Major)
Leave of absence for military service: Feb 2021 - Nov 2022

University of California, San Diego Jan 2026 - Mar 2026
Exchange Student, Computer Science and Engineering *San Diego, CA, USA*
GPA: 4.00 / 4.00

PUBLICATIONS

[P1] TLDR: Unsupervised Goal-Conditioned RL via Temporal Distance-Aware Representations
Junik Bae, Kwanyoung Park, Youngwoon Lee
Conference on Robot Learning (CoRL), 2024

[P2] Exploiting Semantic Reconstruction to Mitigate Hallucinations in Vision-Language Models
Junik Bae*, Minchan Kim*, Minyeong Kim*, Suhwan Choi, Sungkyung Kim, Buru Chang
(*: equal contribution)
European Conference on Computer Vision (ECCV), 2024

[P3] The PokeAgent Challenge: Competitive and Long-Context Learning at Scale
Seth Karten*, Jake Grigsby*, Tersoo Upaa Jr, **Junik Bae**, et al.
(*: equal contribution)
arXiv preprint, 2026

AWARDS AND HONORS

[C1] **1st Place**, PokeAgent Challenge: RPG Speedrunning Track (NeurIPS 2025) Oct 2025 - Nov 2025
Award: $1,500 + $700 (GCP credits), competition report co-authorship
Task: Developing an autonomous Pokemon Emerald speedrun agent for long-horizon decision-making
Method: Voyager-style executable skill generation + RL pipeline

[C2] **1st Place**, 2022 Military AI Competition Nov 2022 - Dec 2022
Awarded by Korean Minister of Science and Technology, ₩20,000,000 (≈ $15,000)
Preliminary Task: Change detection on buildings in aerial image data
Final Task: Image denoising for all-weather operations

[C3] **2nd Place**, 2022 Korean AI Competition Aug 2022 - Sep 2022
Awarded by Korean Minister of Science and Technology, ₩10,000,000 (≈ $7,500)
Task: Speech recognition on Korean dialect speech datasets

[C4] **2nd Prize**, Product Recognition Challenge on Self-service Stand Sep 2021 - Oct 2021
Awarded by Chairman of Electrical and Computer Engineering Department at SNU
Task: High-precision-and-speed object detection on self-service stand images

- **SNU Semiconductor Track Scholarship** Mar 2024 - Present
  Scholarship of ₩10,000,000 (≈ $7,000) to be conferred upon graduation

RESEARCH INTERESTS

My long-term goal is to develop exploration-driven, creative agents that can achieve superhuman performance in open-ended environments. In particular, I am interested in:

- **General unsupervised exploration algorithms** (e.g., [**P1**])
- **Knowledge-guided, sample-efficient exploration** (e.g., [**C1**], [**P3**])
- **LLM-based reasoning and agents** (e.g., [**C1**], [**P2**], [**P3**])

EXPERIENCE

UT Austin Robot Perception and Learning Lab **Visiting Researcher** Apr 2026 - Jun 2026
Hosted by Prof. Yuke Zhu

- Conducting research on meta reinforcement learning with a focus on context management and test-time scaling.

Yonsei RLLAB **Research Intern** Jan 2024 - Present
Advisor: Prof. Youngwoon Lee

- Researched and developed unsupervised goal-conditioned RL algorithm leveraging *temporal distance* for both exploration and goal-reaching [**P1**].
- Built a knowledge-guided RL pipeline using LLM-derived priors for long-horizon exploration, achieving first place in the NeurIPS 2025 PokeAgent Challenge (speedrun track) [**C1**] [**P3**].

SNU Deep Learning Research Club **Member (13th Cohort)** Sep 2023 - Present

- Co-developed framework to mitigate hallucinations in vision-language models using semantic image reconstruction and reinforcement learning [**P2**].

SNU Vision and Learning Lab **Research Intern** Jul 2023 - Sep 2023
Advisor: Prof. Gunhee Kim

- Developed face generation and video scene segmentation pipeline used in MBC entertainment show.
- Researched lifelong evaluation pipelines for retrieval-augmented generation models on dynamic text data streams.

Naver Cloud Speech Synthesis Team **Research Intern** Jan 2023 - Feb 2023

- Designed and implemented text-to-speech (TTS) model tailored for proprietary Korean speech data.

Republic of Korea Air Force **Software Development Specialist** Feb 2021 - Nov 2022

- Developed AI models to detect prohibited items in X-ray images.
- Built ML pipelines for gunshot detection and scoring during shooting training.
- Developed ML forecasting pipeline to predict munitions demand.

PROJECTS

- **TTS Model Implementation. (450+ stars)** (Github link) Implemented *NaturalSpeech: End-to-End Text to Speech Synthesis with Human-Level Quality* by Microsoft, former state-of-the-art model on LJ Speech Dataset. This is the first and only public implementation to the best of my knowledge.